

ArchiMate Modeling Guide

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***Purpose:** Comprehensive guide to ArchiMate modeling across Business, Application, and Technology layers*

What is ArchiMate?

ARCHIMATE OVERVIEW
ArchiMate is an OPEN STANDARD enterprise architecture modeling language maintained by The Open Group (same as TOGAF) PURPOSE: • Describe, analyze, and visualize enterprise architecture • Common language for architects, stakeholders • Works alongside TOGAF ADM KEY FEATURES: • Three main layers (Business, Application, Technology) • Standard notation and symbols • Relationships between elements • Viewpoints for different stakeholders

ArchiMate Core Framework

ARCHIMATE CORE FRAMEWORK		
ASPECTS		
PASSIVE STRUCTURE	BEHAVIOR	ACTIVE STRUCTURE

L A Y E R S	BIZ	Business Object	Business Process Function	Business Actor/Role
	APP	Data Object	Application Service Function	Application Component
	TECH	Artifact	Technology Service Function	Node Device System SW

Layer Colors (Standard)

ARCHIMATE COLOR CODING			
	YELLOW	= Business Layer	(Strategy, Motivation)
	BLUE	= Application Layer	(Applications, Data)
	GREEN	= Technology Layer	(Infrastructure)
	PURPLE	= Implementation Layer	(Projects, Work Packages)
	WHITE	= Physical Layer	(Physical elements)

BUSINESS LAYER (Yellow)

Business Layer Elements

BUSINESS LAYER ELEMENTS	
ACTIVE STRUCTURE (Who/What performs)	
	Business Actor A person or organizational unit



Actor

Example: Customer, Employee, Partner



Role

Business Role
Responsibility for specific behavior
Example: Account Manager, Approver



Collaboration

Business Collaboration
Aggregate of roles working together
Example: Project Team, Review Board

BEHAVIOR (What is performed)



Process

Business Process
Sequence of behaviors achieving result
Example: Order Processing, KYC Verification



Function

Business Function
Collection of behavior based on skill/resource
Example: Payment Processing, Customer Management



Service

Business Service
Service exposed to environment
Example: Account Opening, Loan Application



Event

Business Event
Something that triggers behavior
Example: Order Received, Payment Completed

PASSIVE STRUCTURE (What is acted upon)



Object

Business Object
Concept relevant from business perspective
Example: Order, Invoice, Customer, Product



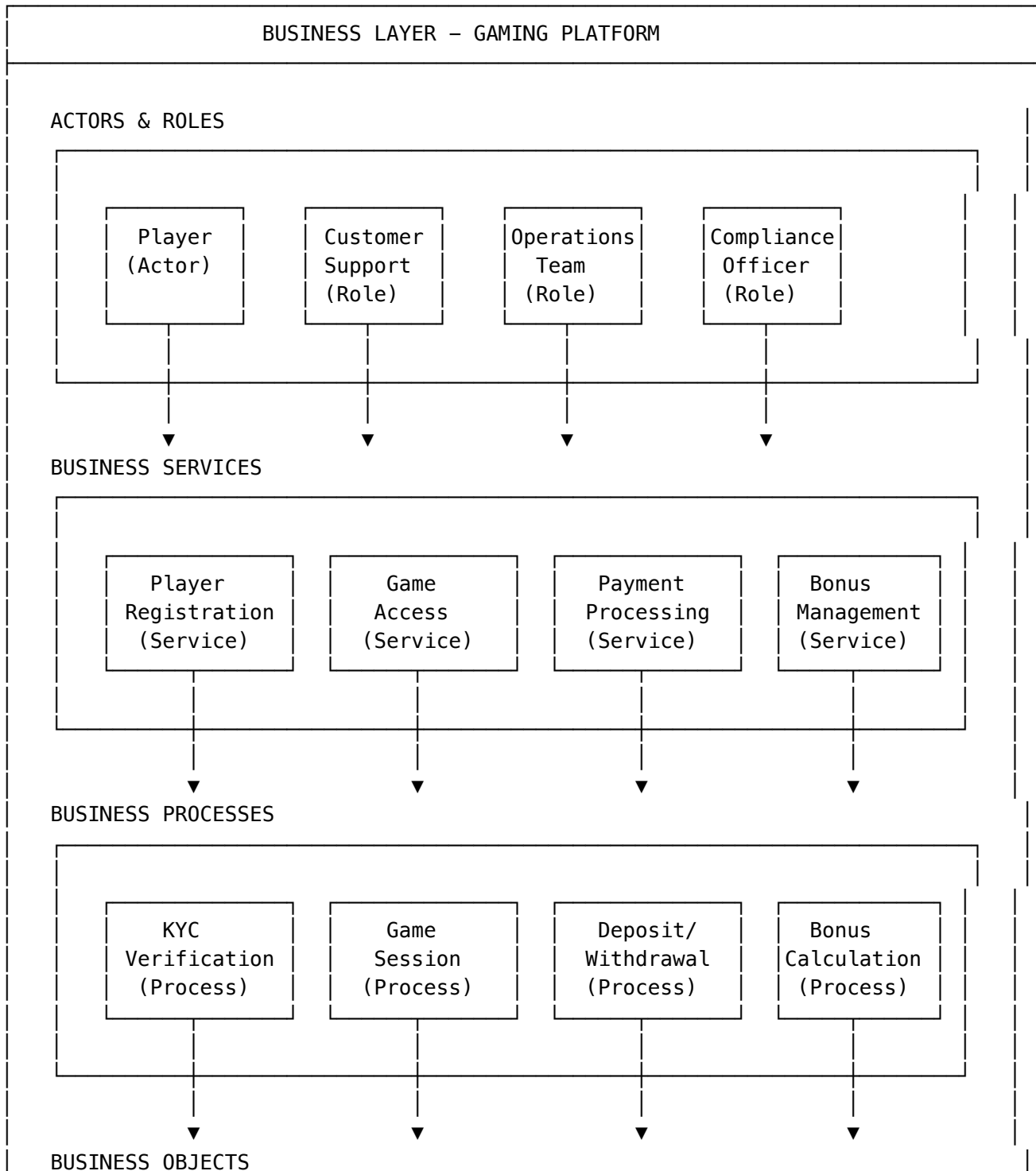
Contract

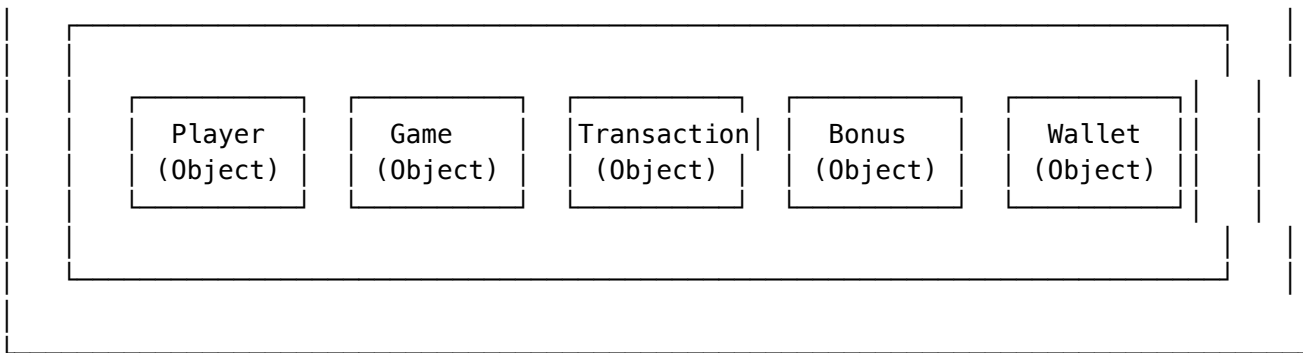
Contract
Formal agreement between parties
Example: SLA, NDA, Service Agreement



Product
Coherent collection of services
Example: Premium Account, Gaming Package

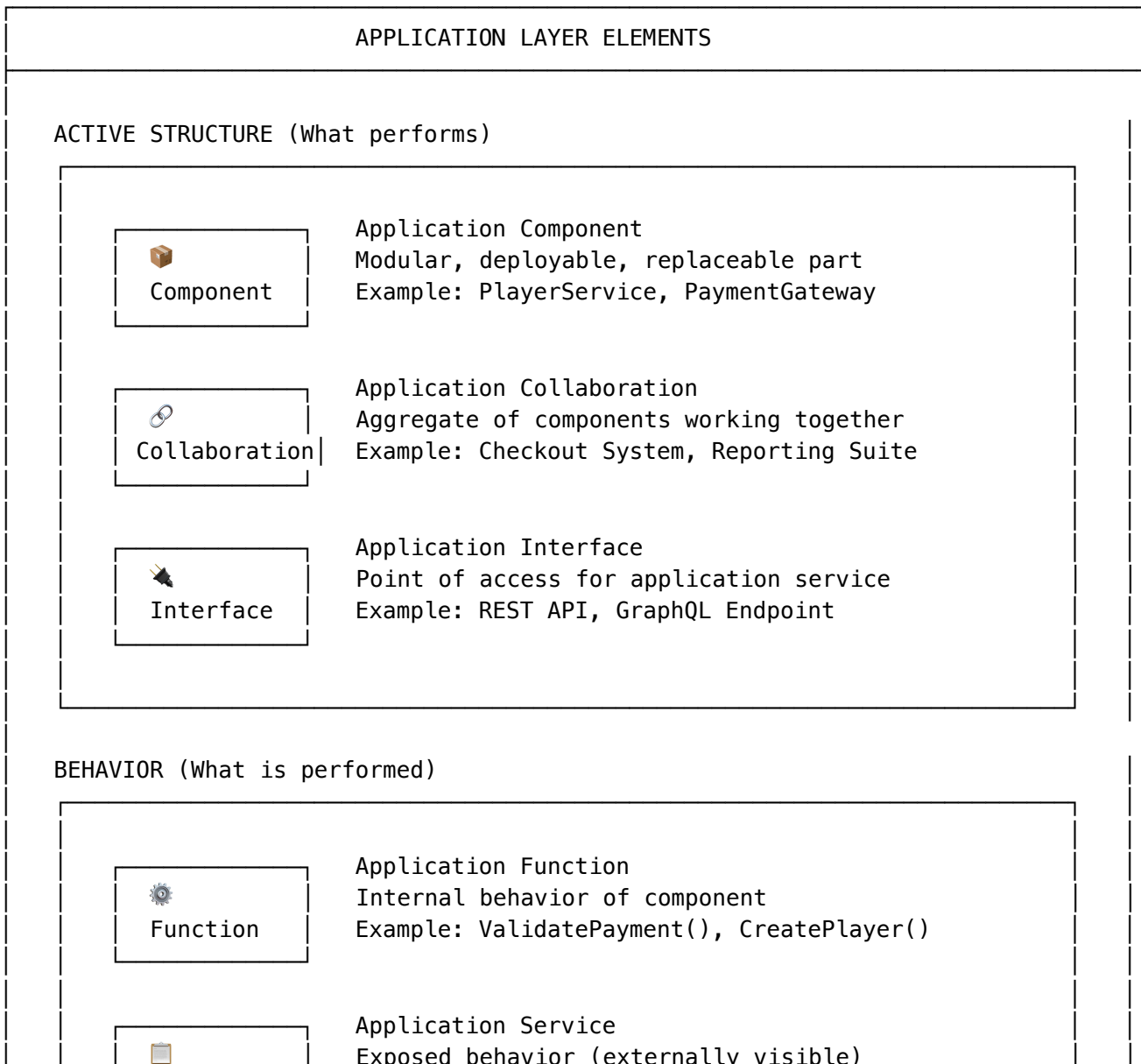
Business Layer Example - Gaming Platform

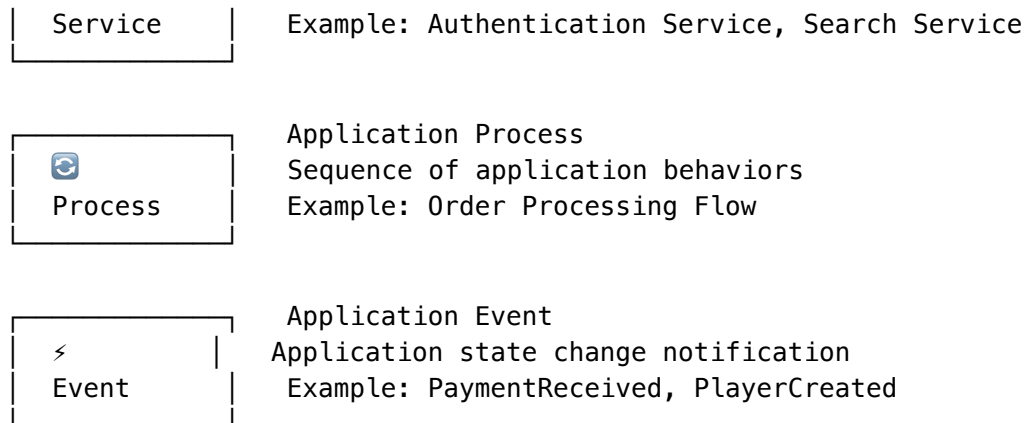




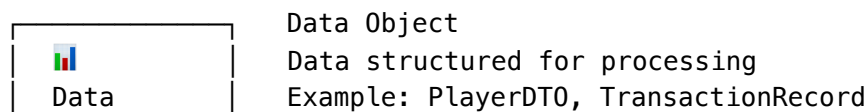
APPLICATION LAYER (Blue)

Application Layer Elements

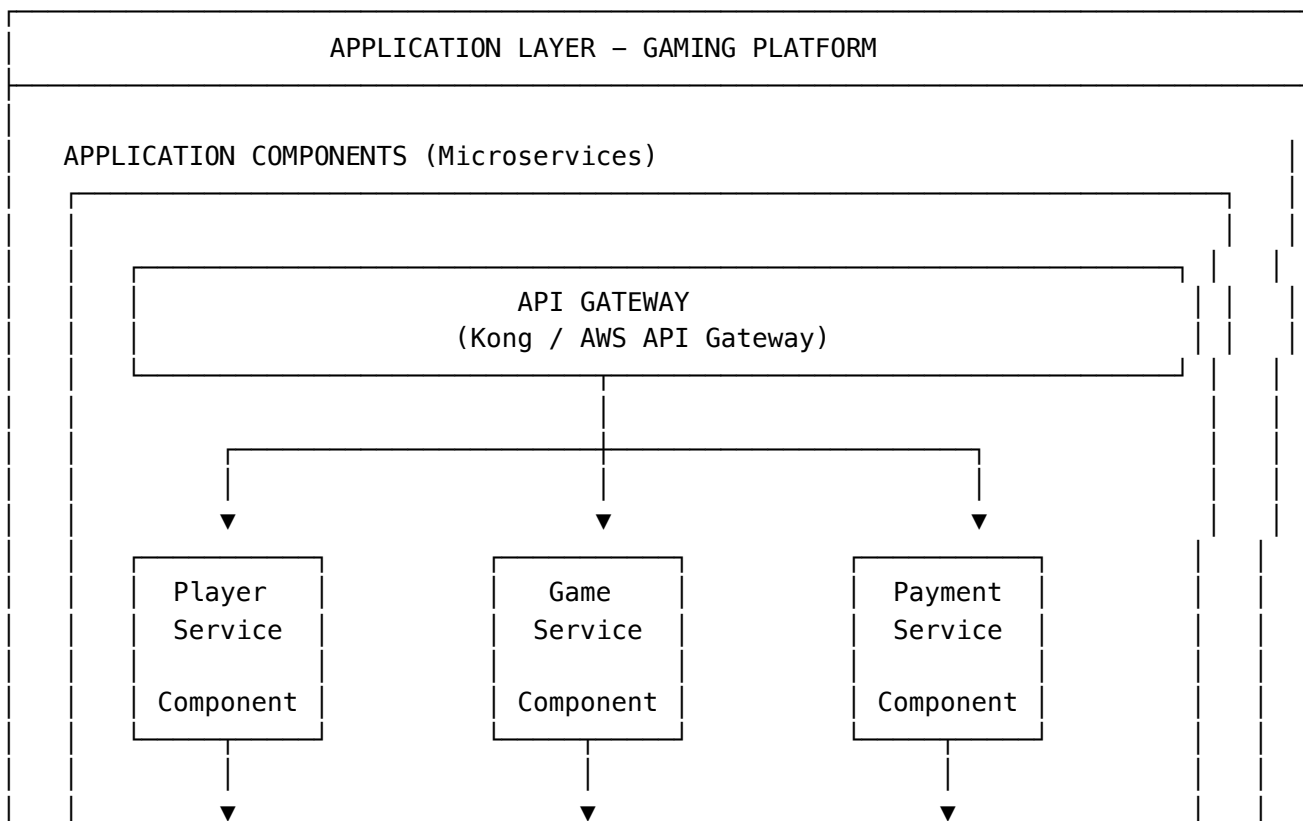


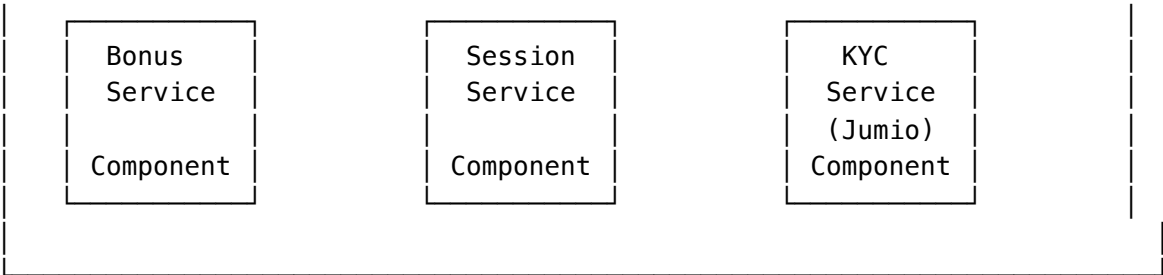


PASSIVE STRUCTURE (What is processed)

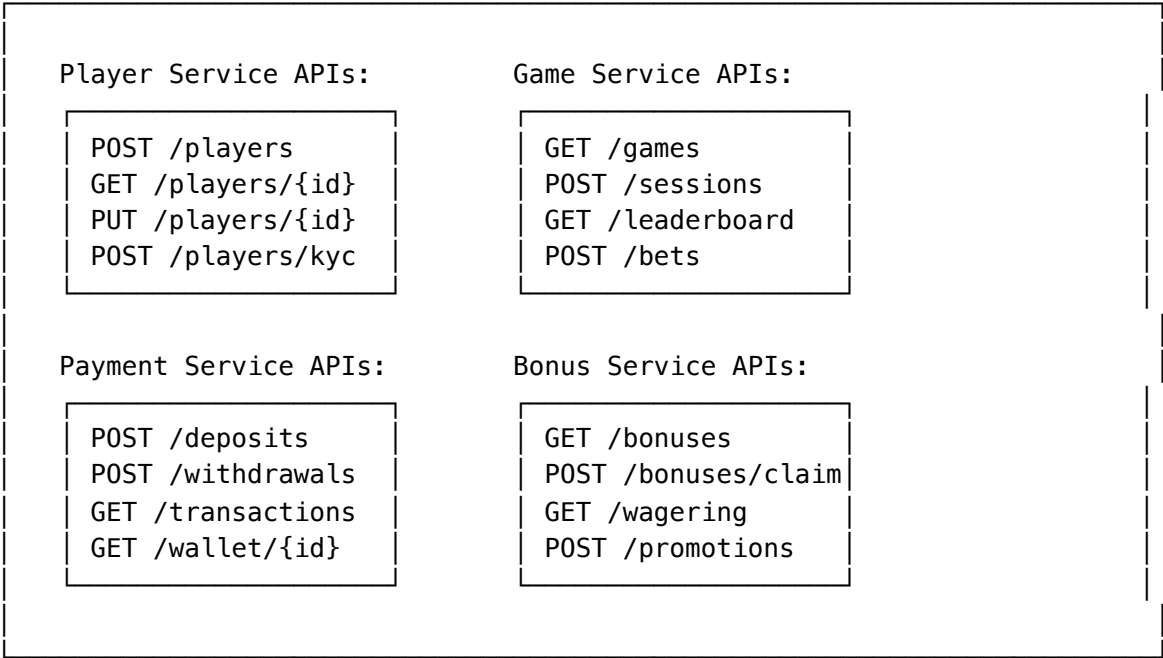


Application Layer Example - Gaming Platform

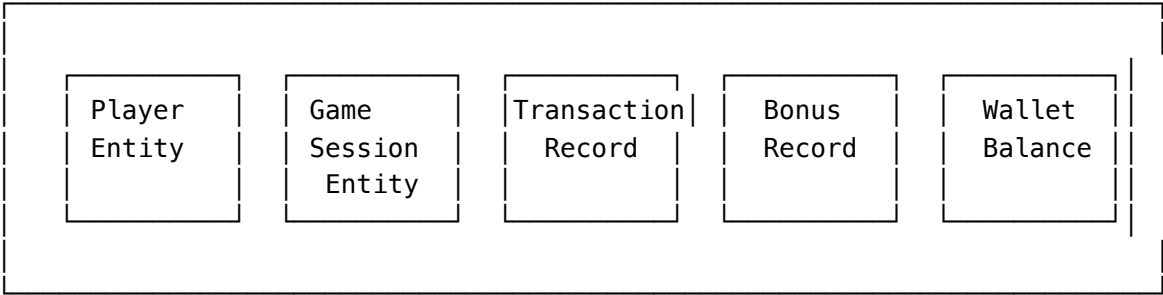




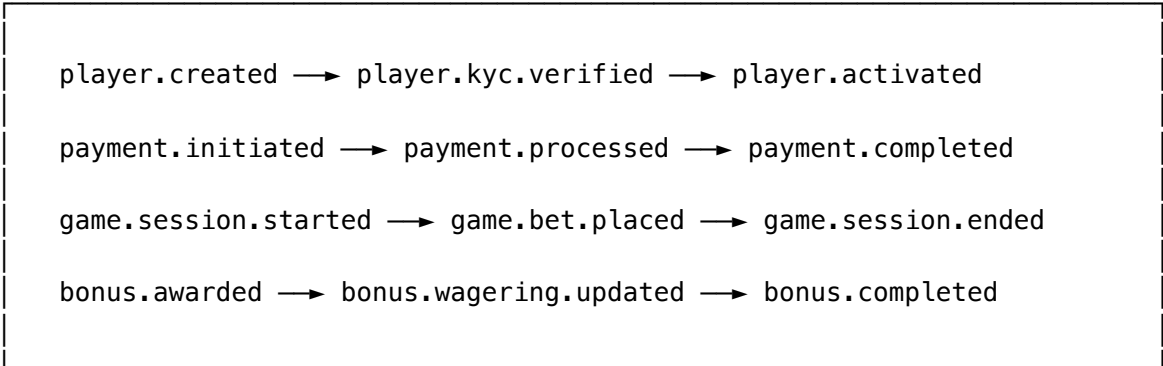
APPLICATION SERVICES (APIs)



DATA OBJECTS

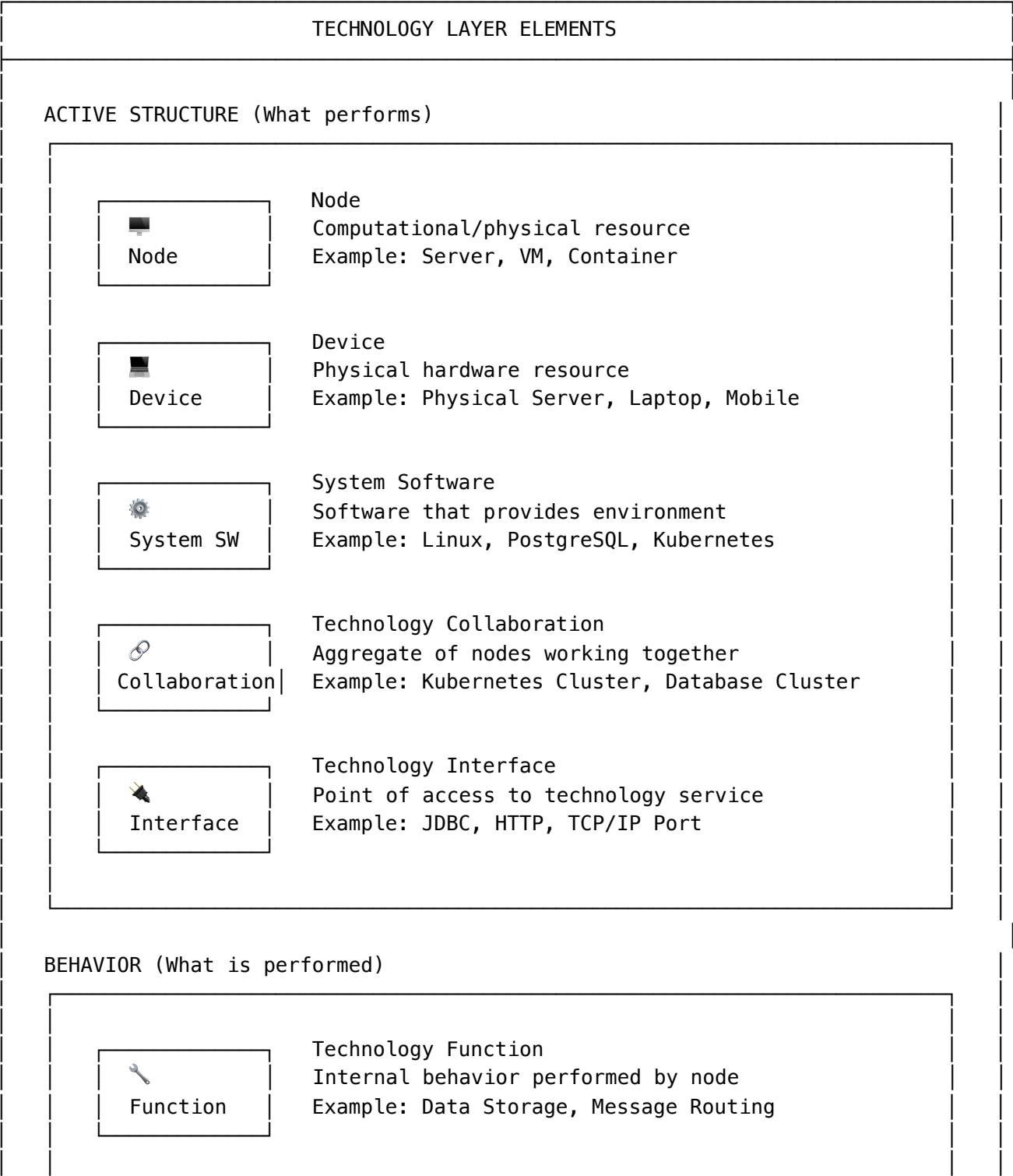


APPLICATION EVENTS (Kafka Topics)



TECHNOLOGY LAYER (Green)

Technology Layer Elements



BEHAVIOR (What is performed)


Function

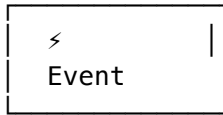
Technology Function
Internal behavior performed by node
Example: Data Storage, Message Routing



Technology Service
Externally visible technology behavior
Example: Database Service, Caching Service



Technology Process
Sequence of technology behaviors
Example: Backup Process, Deployment Pipeline



Technology Event
Technology state change
Example: Server Started, Failover Triggered

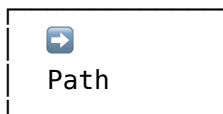
PASSIVE STRUCTURE (What is used)



Artifact
Physical piece of data
Example: JAR file, Docker Image, Config file

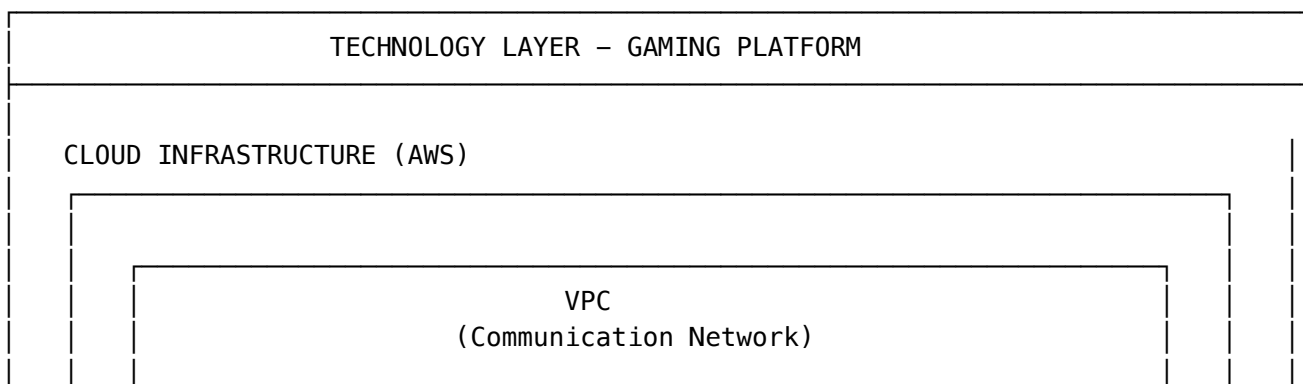


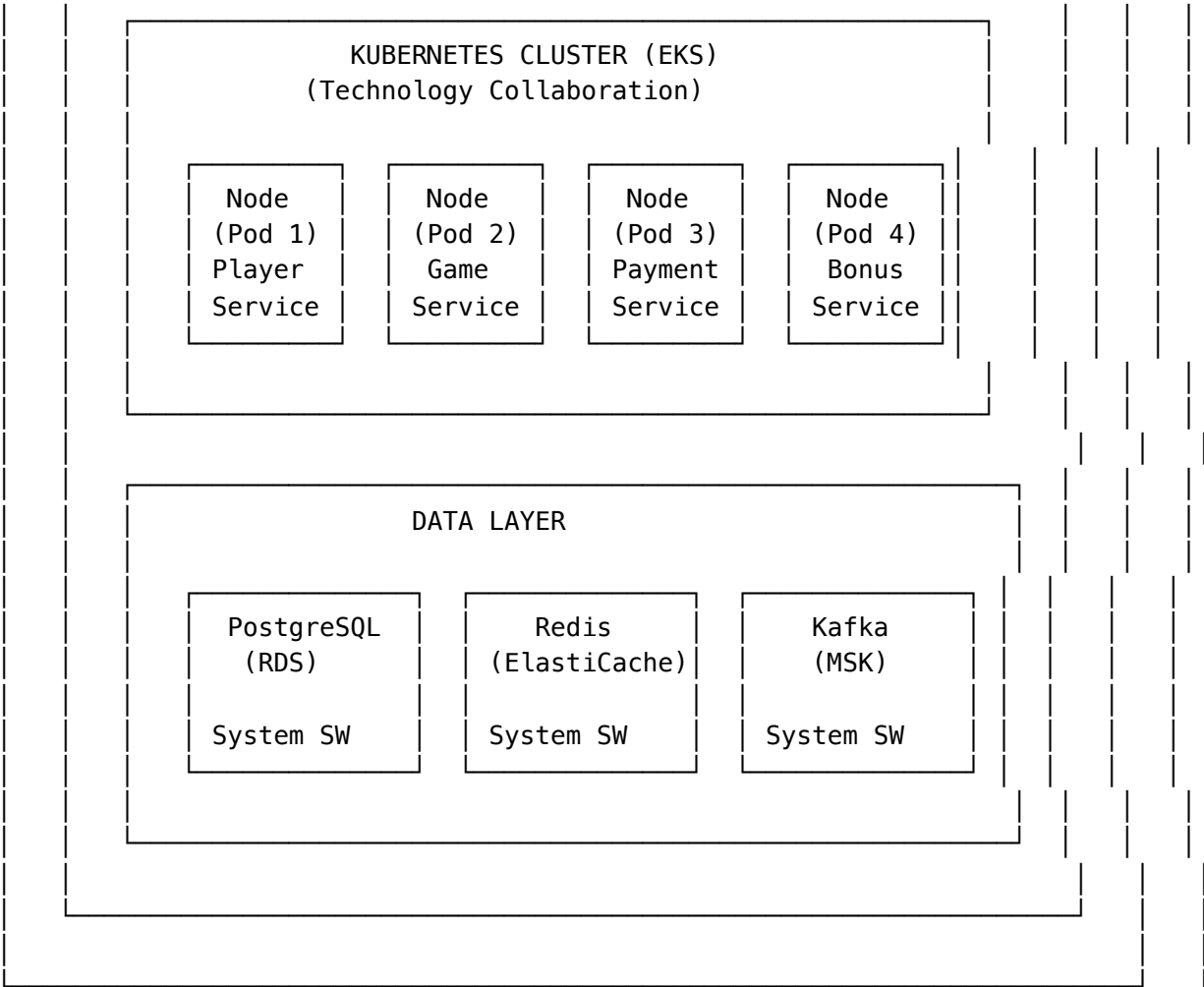
Communication Network
Network for communication between nodes
Example: VPC, LAN, WAN, Internet



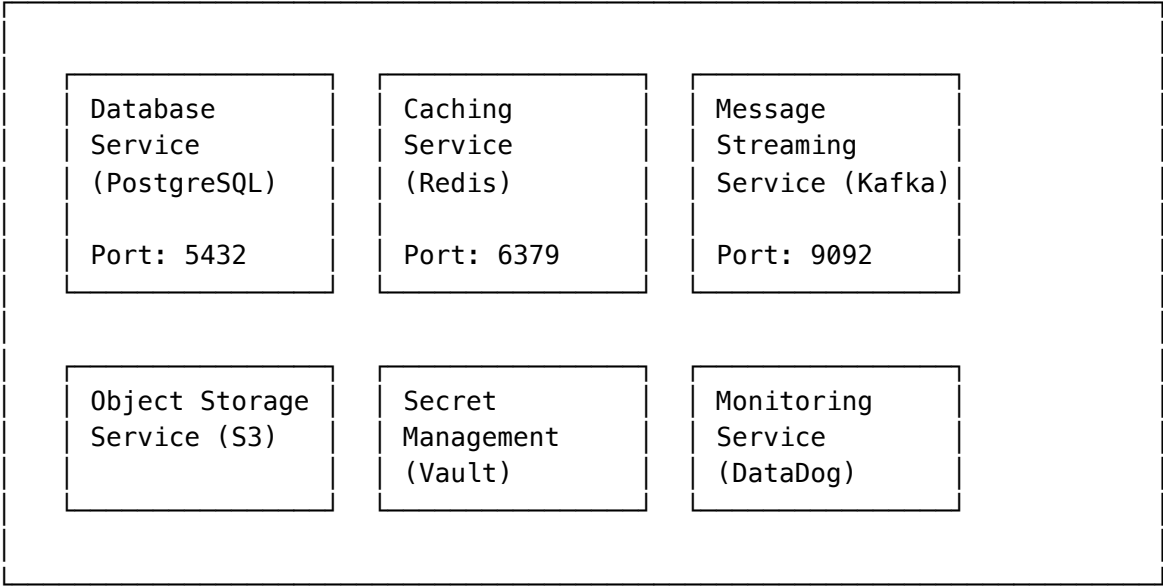
Path
Link between nodes through network
Example: API Route, Network Path

Technology Layer Example - Gaming Platform

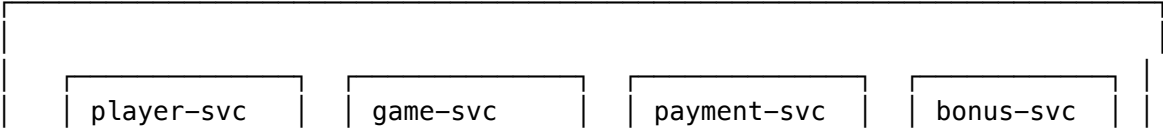


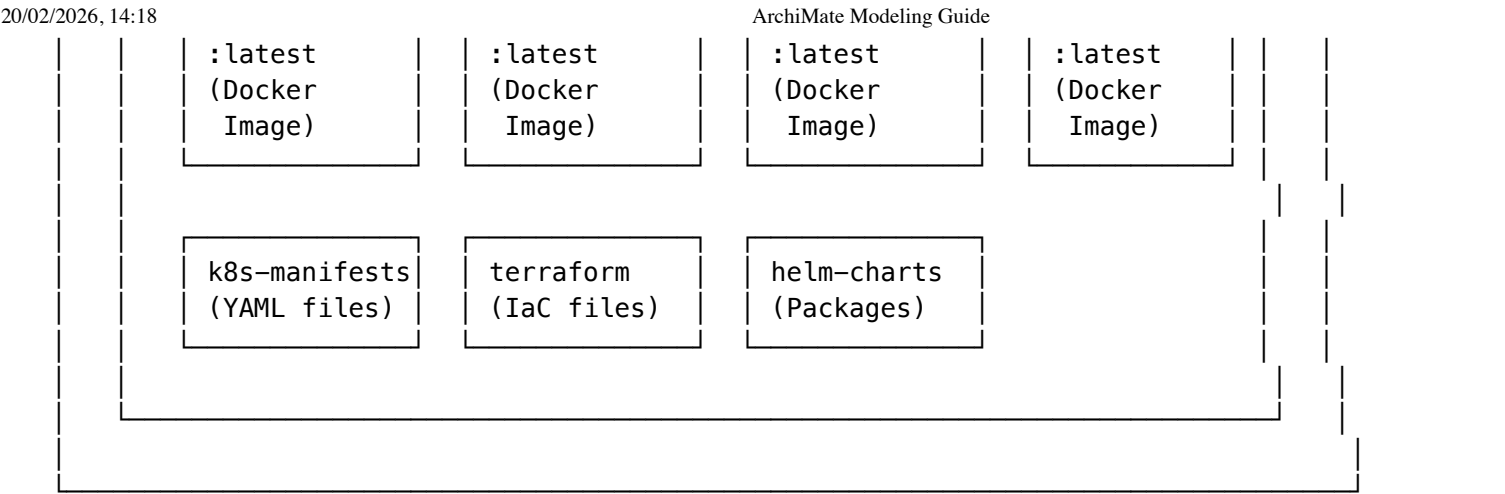


TECHNOLOGY SERVICES



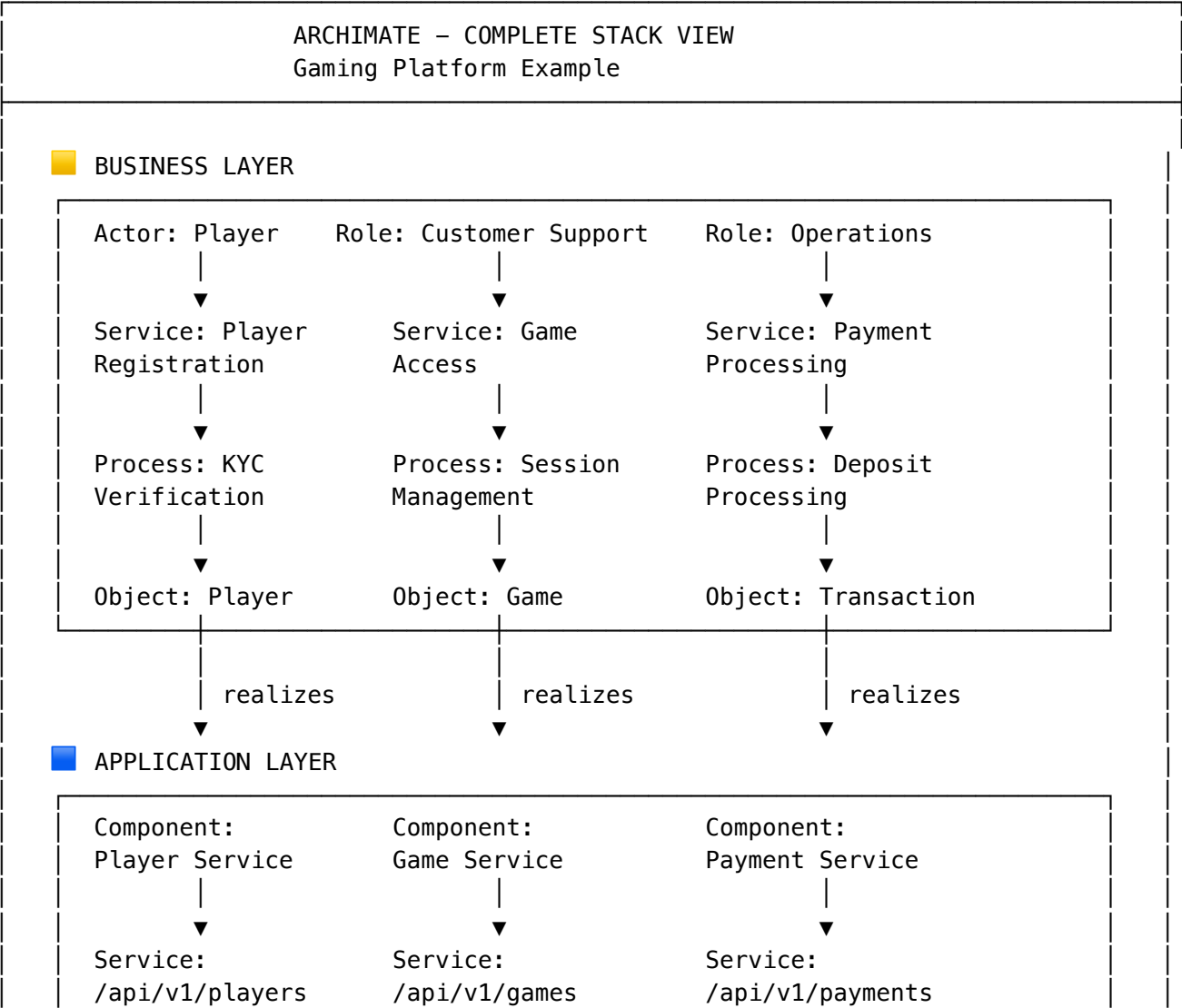
ARTIFACTS (Deployables)

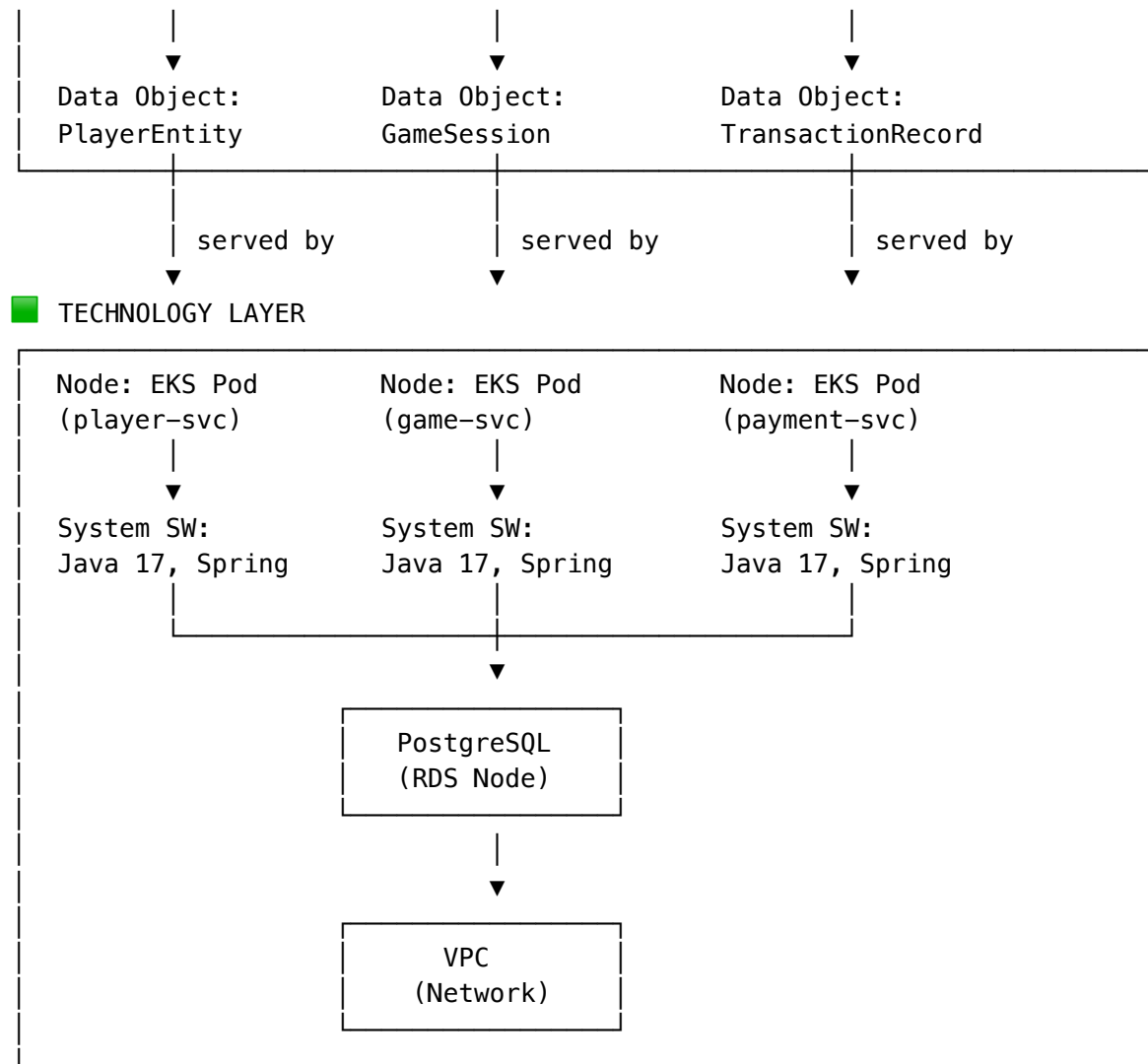




CROSS-LAYER VIEW

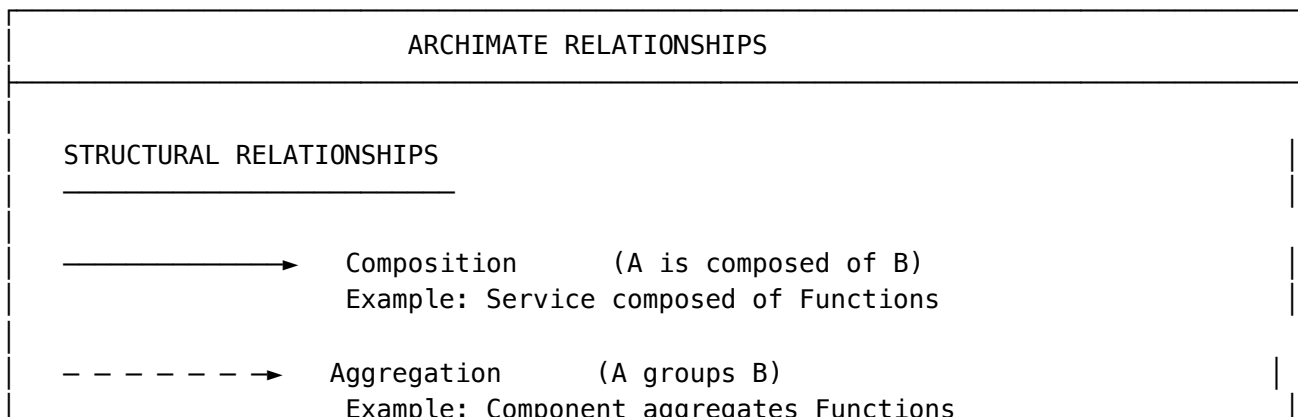
Complete Stack View

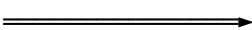
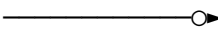




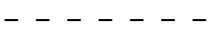
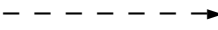
RELATIONSHIPS

ArchiMate Relationships

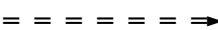


-  Assignment (A performs B, A is assigned to B)
Example: Actor assigned to Role
-  Realization (A realizes B)
Example: Application realizes Business Service

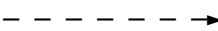
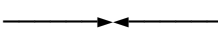
DEPENDENCY RELATIONSHIPS

-  Serving/Used By (A serves B, B uses A)
Example: Application Service serves Business Process
-  Access (A accesses B)
Example: Process accesses Data Object

DYNAMIC RELATIONSHIPS

-  Flow (A flows to B)
Example: Data flows from A to B
-  Triggering (A triggers B)
Example: Event triggers Process

OTHER RELATIONSHIPS

-  Association (General relationship)
-  Specialization (A is specialization of B)
Example: VIP Player specializes Player

VIEWPOINTS

Common ArchiMate Viewpoints

ARCHIMATE VIEWPOINTS	
Viewpoint	Purpose / Audience
Organization	Shows structure of enterprise Audience: Management

Business Process	Shows business processes and flow Audience: Business analysts, Process owners
Application Structure	Shows application components Audience: Application architects
Application Usage	Shows how applications support business Audience: Business & IT stakeholders
Infrastructure	Shows technology infrastructure Audience: Infrastructure architects
Layered	Shows all layers together Audience: Enterprise architects
Service Realization	Shows how services are realized Audience: Service owners, Architects

MAPPING: TOGAF to ArchiMate

TOGAF ADM → ARCHIMATE MAPPING

TOGAF Phase	ArchiMate Layer / Elements
Phase A: Vision	Motivation elements (Goals, Drivers) Strategy elements (Capabilities)
Phase B: Business Architecture	 BUSINESS LAYER Actors, Roles, Processes, Services, Objects
Phase C: Data Architecture	 APPLICATION LAYER (Data Objects) Data Objects, Data Services
Phase C: Application Architecture	 APPLICATION LAYER Components, Services, Interfaces
Phase D: Technology Architecture	 TECHNOLOGY LAYER Nodes, Devices, System Software, Networks
Phase E & F Opportunities & Migration	 IMPLEMENTATION LAYER Work Packages, Deliverables, Plateaus

TOOLS

ArchiMate Modeling Tools

ARCHIMATE TOOLS

FREE / OPEN SOURCE:

- Archi (archimatetool.com) – Most popular free tool
- Draw.io – With ArchiMate stencils
- Modelio – Open source modeling

COMMERCIAL:

- Sparx Enterprise Architect
- BiZZdesign Enterprise Studio
- LeanIX
- MEGA HOPEX
- Orbus iServer
- Avolution ABACUS

Summary

ARCHIMATE SUMMARY

THREE MAIN LAYERS:

-  Business Layer – WHO does WHAT (Actors, Processes, Services)
-  Application Layer – WHAT applications support business
-  Technology Layer – HOW applications are deployed

KEY CONCEPTS:

- Active Structure = WHO/WHAT performs (Actors, Components, Nodes)
- Behavior = WHAT is done (Processes, Services, Functions)
- Passive Structure = WHAT is acted upon (Objects, Data, Artifacts)

RELATIONSHIPS:

- Realization = A implements/realizes B
- Serving = A serves/supports B
- Assignment = A is assigned to B
- Flow = Data/information flows from A to B

USE WITH TOGAF:

- ArchiMate is the MODELING LANGUAGE
- TOGAF ADM is the PROCESS
- Together they provide complete EA approach